
Introduction to China's Financial System

LECTURE 9 DERIVATIVES MARKETS

Two basic concepts

- Primary financial instruments: also called basic financial instruments, a standard financial investment that often trade on mainstream exchanges. Their price is based directly on market value and determined based on assumptions about their individual characteristics. The main function is to mediate the transformation of savings into investment or used for debt repayment certificates.
 - Primary instruments include cash-traded products like stocks, bonds, currencies, and spot commodities, etc.
- Derivative financial instruments, also known as derivative securities and derivative products, refer to financial instruments whose value depends on the price of the underlying asset (primary financial instruments), its main function is to manage the risk exposure related to the original financial instrument.
 - Derivative instruments include: forward, futures, options and swaps

Derivative financial instruments

- Since the 1970s, there have been two great revolutions in the world.
 - The information technology revolution centered on computer and communication technology.
 - The financial revolution centered on financial innovation. The innovation of financial instruments with Derivative Securities such as futures and options is the core of this financial revolution.
- The background of derivative securities: the disintegration of Bretton Woods System in the early 1970s, the fluctuation of interest rate and exchange rate led to the instability of financial asset prices.
- Derivative securities are the financial instruments with the shortest history, but the fastest development and the largest trading volume in the world.

Characteristics of derivative securities

- Advantages
 - to replace uncertainty with certainty, control the risk caused by interest rate and exchange rate fluctuations.
- Disadvantages
 - the inherent leverage and virtuality increase the financial risk, speculation has become a common phenomenon.

Functions of derivative securities

- A derivative security is a contract in which the transaction is a zero-sum game (considering commissions, it is actually negative--net outflow of assets--for both buyers and sellers).
- The trading of derivative securities is actually a redistribution of risk, it does not create wealth.
- Derivatives are highly leveraged. This will not only increase the risk of the market, but also reduce the transaction cost and improve the liquidity of the market.

Types of derivative securities

- Forward Contracts
- Futures Contracts
- Options
- Swaps

Forward contracts

- Forward contracts are contracts in which two parties agree to buy or sell a certain amount of assets at a given price at a future date.
 - Forward contracts arise from the need to avoid the risk of spot trading and are non-standard contracts.
 - Before signing a forward contract, the two parties can negotiate the details of the delivery place, delivery time, delivery price, contract size, quality of the subject matter, so as to meet the needs of both parties as much as possible.
- The main advantage of a forward contract is "flexibility".
- Forward contracts include forward interest rate agreements, forward foreign exchange contracts and forward stock contracts, among others.
- Forwards are mainly conducted through the over-the-counter market.

Futures contracts

- Futures contracts are standardized agreements in which the parties agree to buy or sell a standard quantity of a financial instrument on an agreed future date and under agreed conditions, including price, place of delivery, and method of delivery.
- A futures contract is a standardized forward contract. The contract size, delivery date and delivery location of futures contracts are standardized.
- The price specified in the contract is the Futures Price.
 - A futures price is simply the forward price of a standardized forward contract.

Characteristics of futures contracts

- Futures contracts are conducted on exchanges. The two sides of the transaction do not have direct contact with each other, but settle with the clearing department of the exchange or a special clearing company.
- A buyer or seller of a futures contract may terminate a futures position (i.e., liquidate a position 平仓) by taking a hedging transaction prior to the delivery date without a final physical delivery.
- Futures trading is settled on a daily basis, and both buyers and sellers must open a special margin account with a brokerage firm before trading.

Types of futures contracts

- Depending on the subject matter, financial futures can be divided into:
 - **Interest rate futures**: in which the price of the underlying asset (an interest-bearing instrument such as bill, note, bond) depends on the level of interest rates.
 - **Stock index futures**: in which the underlying asset is a stock price index.
 - **Foreign exchange futures**: the underlying asset is foreign exchange.

Financial options

- A financial option refers to a financial derivative that, the buyer of the option after paying a certain premium, has the right (but no obligation) to buy or sell some underlying assets in the future at a pre-agreed price within a specified period.
- The agreed price is called “Striking Price” or “Exercise Price”.

Classification of financial options

- By ways of exercising
 - **American option**: an option in which the purchaser of the option can exercise the right on or at any time prior to the expiration date.
 - **European option**: an option can only be exercised on the expiration date.
- By the nature of the option
 - **Call option**: the buyer of an option has the right to buy the product at an agreed price at an agreed time in the future for a certain premium.
 - **Put option**: the buyer of an option has the right to sell the product at an agreed price for a certain period of time in the future.

Classification of financial options

- By the subject matter (underlying assets)
 - **Stock option:**
 - the right to buy or sell a certain number of related shares at the agreed price on or before the expiration date specified in the contract.
 - **Stock index option:**
 - the buyer has the right to settle the profit and loss between the agreed index and the actual market index within the validity period or upon expiration of the contract.
 - **Interest rate option:**
 - the buyer obtains the right to buy or sell a certain amount of interest rate instrument at a certain interest rate (price) during the validity period or expiration of the contract.
 - **Currency option:**
 - the buyer obtains the right to purchase or sell a certain amount of foreign exchange assets at the agreed exchange rate within the validity period or upon expiration of the contract.
 - **Financial futures contract option:**
 - it gives its holder the right to buy and sell a specific financial futures contract at an agreed price within a specified time.

Swaps

- Swap refers to a contract in which two parties (or more parties) sign a swap agreement to exchange cash flows or liabilities of different financial instruments within a future agreed period.
 - Swap is a kind of financial transaction behavior in which different debts, debts with different interest rates, or debts with different delivery periods, are exchanged between the two parties according to the market conditions, and a series of payments are made within the agreed period.
- Essence: it is a group of forward contracts; The parties to the swap are usually mutually beneficial.
- Swaps are mainly conducted through the over-the-counter market.

Types of swaps

- Interest rate swaps
 - An interest rate swap is when two parties agree to exchange cash flows based on the same notional principal of the same currency for a certain period of time in the future, with one party's cash flows calculated based on a floating rate and the other party's cash flows calculated based on a fixed rate.
 - The maturity of the swap is usually over 2 years to 15 years.
- Currency swaps
 - A currency swap is an exchange of principal and fixed interest in one currency for an equivalent principal and fixed interest in another currency.
 - It helps to avoid foreign exchange risk and reduce the financing cost of both sides.
 - It enables traders, such as governments and enterprises, to take advantage of foreign capital markets and obtain foreign currency funds.

Example: Interest rate swap

- Suppose that
 - Company A can obtain a dollar loan at a 5% fixed rate or a 4% floating rate;
 - Company B can obtain a dollar loan at a 7% fixed rate or a 4.5% floating rate loan.
- At present,
 - Company A obtains a loan with a floating interest rate of 4%.
 - Company B obtains a fixed interest rate of 7%.
- The two companies have agreed that after Company B bears the interest rate of 1% to Company A, the two parties will exchange the obligation of interest payment as following:
 - Company A will pay the floating interest for Company B;
 - Company B will pay the fixed interest for Company A.
- Results of the swap:
 - The total interest rate paid by Company B then is $1\% + 5\% = 6\%$
 - The floating interest rate paid by Company A on behalf of Company B is 4.5%, but since Company B paid Company A 1% interest rate in advance, the total interest rate actually paid by Company A is 3.5%.
- Significance: The two companies effectively reduce their interest costs through an interest rate swap
 - Company A paid a lower effective floating rate (3.5%) than it should have paid (4%)
 - The actual fixed rate paid by Company B (6%) is lower than what it should be paying (7%)

Example: Currency swap

- Suppose that
 - the current exchange rate between the UK pound and the US dollar is 1.35.
 - Company A of US borrowed 1 million pounds at a fixed interest rate of 2% on the Euro money market;
 - Company B of UK borrowed 1.35 million US dollars at a fixed interest rate of 3% .
- It is agreed that, during the term of the debt,
 - Company B of UK will provide pounds to Company A of US to repay the interest;
 - Company A of US will provide US dollars to Company B of UK to repay the interest.
 - Company B should make a net payment of $(1.35m \cdot 3\% - 1.35m \cdot 2\%) = 13.5$ thousand USD to Company A.
- When debts mature, the two parties will pay each other the principal of debt to end the swap.
 - Company B of UK will pay 1 million pounds to Company A of US;
 - Company A of US will pay 1.35 million US dollars to Company B of UK.

Functions of swaps

- Financial swaps can be used to manage interest rate risk and exchange rate risk in the portfolio of assets and liabilities.
- As an off-balance-sheet business, financial swaps can evade foreign exchange control, interest rate control and tax restrictions.

Derivatives market

- A derivatives market is a market in which financial derivatives are traded.
- By trading products, the financial derivatives market can be divided into:
 - Futures market
 - Options market
 - Forward market
 - Swap market
- By trading modes, the financial derivatives market can be divided into:
 - Floor Market (exchange market)
 - Over-the-Counter Market (OTC)

The way derivatives markets are traded

- Floor trading in an exchange
 - also known as a physical market. In terms of the types of trading, futures and the vast majority of options trading belong to the floor trading, with standardized contracts, and, when the trading contract has not expired, traders can easily hedge and close positions.
- Over-the-Counter trading outside the exchange
 - (hereinafter referred to as OTC trading). It is mainly negotiated by the two parties, and the financial derivatives traded are mainly derivatives not listed on the exchange. In terms of the types of transactions, forwards, swaps and some special types of options are often traded OTC.
- In terms of transaction size, OTC transactions account for about 70%, while exchange transactions account for about 30% in China.

Key players of the derivatives market

- According to the different purpose of trading:
 - **Hedgers:** are called risk hedgers, who engage in derivatives trading in order to reduce future uncertainty and reduce or even eliminate risks.
 - **Speculators:** investors who trade in two or more markets at the same time to make a profit, taking advantage of pricing differences in different markets.
 - **Broker:** acts as an intermediary in a transaction for the purpose of facilitating the transaction and receiving a commission.
- From the perspective of market participants, there are buyers and sellers.
 - **Buyers:** include public funds, commercial banks, asset management companies, etc. They are the driving end of market demand and play an important role in the price-determination process.
 - **Sellers:** mainly are securities companies or investment banks. They process or customize derivatives according to the needs of buyers and help buyers to realize risk hedging or establish speculative positions. The seller obtains income by dynamically replicating the income of derivatives and locking the price difference.

China's Derivatives Market

- Participants of China's derivatives market
 - Exchanges and trading platforms
 - Investors and dealers
 - Regulators
- Development of China's derivatives market
- Current situation of China's derivatives market

An overview of China's derivatives market

- At present, China's financial derivatives market can be divided into
 - **Futures exchange market:** It is an exchange trading market supervised by China Securities Regulatory Commission. It is composed of four commodity futures exchanges, and one financial futures exchange.
 - Four commodity futures exchanges: Shanghai Futures Exchange, Zhengzhou Commodity Exchange, Dalian Commodity Exchange, Guangzhou Futures Exchange.
 - One financial futures exchange: China Financial Futures Exchange.
 - **OTC financial derivatives market:** it mainly includes
 - Inter-bank OTC derivatives market (which is supervised by the PBoC)
 - Bank-counter OTC derivatives market (which is supervised by the PBoC and NFRA)
 - Securities firms OTC derivatives market (which is supervised by CSRC)

The development of China's futures market

- On October 12, 1990, Zhengzhou Grain Wholesale Market was established with the approval of The State Council. Based on spot trading, it introduced futures trading mechanism, **marking the first step in the development of China's futures market.**
- After more than 30 years of development, it has experienced five stages:
 - The initial development stage
 - The cleaning and rectification stage
 - The normative development stage stage
 - The innovation and exploration stage
 - The high-quality development stage

The development of China's futures market

- Initial development stage (1990-1993)

- In the early 1990s, there were more ideological obstacles to the development of futures markets. Therefore, it was only possible to declare a wholesale market, and the financial attribute (T+0 trading) was hidden in the trading mechanism.
- In October 1990, Zhengzhou Grain Wholesale Market was officially opened, which was the first market with futures trading varieties in China, but it was not called futures exchange at that time.
- In 1992, after the 14th National Congress of the Communist Party of China, the futures market developed rapidly.
 - On January 18, 1992, Shenzhen Non-ferrous Metal Exchange was established and officially opened.
 - On May 28, 1992, Shanghai Metal Exchange was opened.
 - In September 1992, Guangdong Wantong Futures Brokerage Co., LTD., China's first futures brokerage company, was established. Subsequently, China International Futures Corporation was established.
- By the end of 1993, in just over a year, there were more than 50 futures exchanges, more than 300 futures brokerage companies and more than 2,000 parallel futures institutions (banks, securities companies and trust companies) in China.
- Due to the “quick work”, a series of problems then have been exposed: serious duplication of trading varieties, serious excessive speculation and frequent market manipulation; The operation of futures trading institutions is not standardized.
- **In particular, the regulatory body of the futures market is not clear**, and various problems arising from the pilot exploration stage need to be rectified urgently.

The development of China's futures market

- Cleaning and rectification stage (1993-2000)

- During this period, **a large number of futures exchanges and futures brokerage companies appeared** in China, including more than 40 exchanges, 500 futures brokerage companies, more than 50 listed trading varieties. Meanwhile, overseas futures and underground futures of futures brokerage companies flooded blindly.
- In November 1993, China Securities Regulatory Commission (**CSRC**) **was set as the futures regulatory agency**, had centralized and unified supervision over the futures market, which marked the beginning of the first round of governance and rectification.
- On May 30, 1994, The State Council issued Document No. 69, which carried out the first major clean-up of the futures industry. After rectification, there were only 15 futures exchanges.
- **On May 18, 1995, Treasury bond futures were stopped by CSRC due to the “3.27 incident”.**
- In 1998, The State Council issued the Notice on Further Rectification and Regulation of the Futures Market, which further regulated the futures market. The number of exchanges changed from 15 to 3, and the number of trading products decreased from 35 to 12.
- In September 1999, the Interim Regulations on Futures Trading were formally implemented, the minimum registered capital of futures brokerage companies was raised to 30 million yuan.
- In December 2000, the China Futures Association was established, playing a role in industry self-discipline management.
- During the rectification process, a large number of futures exchanges and brokerage companies that did not meet the standards were closed down. Some investors suffered losses and market confidence was somewhat affected.

"327 Incident"

- 327 is the code name of the Treasury bond futures contract, which corresponds to the 3-year Treasury bonds issued in 1992 and due in June 1995. The total amount of the bonds issued is 24 billion yuan, and the redemption method is 8% coupon plus discount.
- On February 23, 1995, Shanghai Wanguo Securities Company, the main short side, issued 10.56 million sell orders in the last 8 minutes, with a face value of **211.2 billion yuan**, while the total amount of 327 Treasury bonds was **only 24 billion yuan**, pushing the price of "327" contract from 151.30 to 147.50 yuan, which was confirmed as malicious violation. Later, the Treasury bond futures died due to the "327 Treasury Bond incident". The Financial Times called it "the darkest day in the history of Chinese securities".
- ✓ A bitter price
 - After the incident, on April 25 1995, Wanguo held a board meeting and Chairman Xu Qingxiong and President Guan Jinsheng both resigned. In January 1997, Guan Jinsheng was sentenced to 17 years in prison by the Shanghai High Court.
 - On September 15 1995, Wei Wenyuan, the first general manager of the Shanghai Stock Exchange, also announced his resignation and left the securities market.
 - In April 1996, Wanguo merged with Shenyin Securities Co., LTD., its strongest competitor in the Shanghai market.

Post- “327 Incident” regulation

In 1995, the CSRC suspended the trading of treasury bond futures and reorganized and merged several futures exchanges, which had a significant impact on the pattern of the entire derivatives market.

- Suspension of Treasury Bond Futures Trading
 - In 1995, in order to rectify the disorderly market order and prevent financial risks, the CSRC decided to suspend the trading of treasury bond futures. The implementation of this policy caused this important on-exchange derivatives trading variety at that time to temporarily withdraw from the market, which had a major impact on the entire financial derivatives market.
- Reorganization and Merger of Futures Exchanges
 - In the same year 1995, the CSRC also reorganized and merged futures exchanges, retaining only three commodity exchanges, namely the Shanghai Futures Exchange, the Dalian Commodity Exchange, and the Zhengzhou Commodity Exchange. This further standardized the trading venues of the futures market and strengthened the centralized management of the futures market.

The development of China's futures market

- Normative development stage (2000-2010)
 - China's futures market has stepped into the stage of standardized development since 2000.
 - In December 2000, China Futures Industry Association was formally established.
 - In March 2004, The State Council clearly stipulated that futures brokerage companies belong to financial institutions. Before that, futures companies belonged to the service industry.
 - **In 2006, the China Financial Futures Exchange was established.**
 - It successively listed stock index futures in 2010 and treasury bond futures in 2013, improving China's futures market product system.
 - **In 2006, the China Futures Margin Monitoring Center was established.**
 - China Futures Margin Monitoring Center was renamed as China Futures Market Monitoring Center Co., LTD. on April 15, 2015. It is a non-profit corporate legal person established with the consent of The State Council and the decision of CSRC.
 - By then, China's futures market had formed a "five-in-one" supervision system.
 - In April 2010, The Shanghai-Shenzhen 300 stock index futures contract (CSI 300 Index Futures) was listed, becoming China's first financial futures.

The development of China's futures market

- Innovation and exploration stage (2010-2020)
- Since 2010, China's futures market has entered a critical period of transformation from quantitative expansion to qualitative improvement.
 - On April 16, 2010, China launched the first domestic stock index futures contract-CSI 300 Index Futures.
 - On August 19, 2011, the investment consulting business of the futures company was officially approved.
 - On November 21, 2012, the asset management business of the futures company was officially approved.
 - In 2013, China Futures Industry Association examined and approved the filing applications of 20 futures companies to establish risk management subsidiaries.
 - On September 6, 2013, Treasury bond futures were re-listed.
 - On February 9, 2015, the Shanghai Stock Exchange launched SSE 50 ETF options, enriching the product system of China's financial derivatives market.
 - In 2016, sugar and soybean meal options were listed, achieving a breakthrough in commodity options.
 - In March 2018, China's first internationalized variety, crude oil futures, was listed and traded, taking a new step in the internationalization of the futures market.
 - In 2018, the "Administrative Measures for Foreign-Invested Futures Companies" was officially released and implemented, taking an important step in the opening up of the futures market.

The development of China's futures market

- High-quality development stage (2020-Present)
- China's economy has entered a stage of high-quality development, putting forward higher requirements for the quality and efficiency of financial market services for the real economy.
- With the implementation of the "Law of the People's Republic of China on Futures and Derivatives", the development and supervision of the futures market have entered a new track of legalization.
 - In December 2020, China Securities Commodity Index Co., Ltd. was registered and established
 - In January 2021, Guangzhou Futures Exchange was approved to be established, further optimizing the overall layout of China's futures market and product system.
 - On April 20, 2022, the "Law of the People's Republic of China on Futures and Derivatives" was reviewed and passed, providing a solid legal guarantee for the market development.
 - In December 2022 and July 2023, it respectively launched two new energy futures varieties, industrial silicon and lithium carbonate, providing effective risk management tools for relevant industrial enterprises.
- Since 2022, the market scale has been growing steadily, the variety system has been further improved, the legal construction has been continuously strengthened.

Five futures exchanges in mainland China

- Shanghai Futures Exchange
 - It officially started operations in December 1999. It mainly deals with **non-ferrous metals, precious metals**, etc.
- Dalian Commodity Exchange
 - It was established on February 28, 1993, focusing on **agricultural products and industrial products**.
- Zhengzhou Commodity Exchange
 - It officially launched standardized futures contracts on May 28, 1993. It has characteristics in **agricultural products and chemical products**.
- China Financial Futures Exchange
 - It was established in Shanghai on September 8, 2006. It mainly conducts **financial futures trading**. At present, the main varieties are stock index futures and treasury bond futures.
- Guangzhou Futures Exchange
 - It was inaugurated on April 19, 2021. It is the fifth futures exchange in China. It focuses on futures varieties related to **green development**, including carbon emission rights, and other industrial characteristic varieties closely related to green and low-carbon development.

A brief history of futures products

- Futures are divided into: commodity futures, Treasury bond futures, stock index futures and so on.
- 1. Commodity futures
 - On October 27, 1990, China introduced the futures trading mechanism of the national wholesale market -- China Zhengzhou Grain Wholesale market opened, mainly spot, **the first introduction of futures trading mechanism.**
 - On November 26, 1990, the Shanghai Futures Exchange was established. There are 11 futures trading varieties, including gold, silver, copper, aluminum, zinc, lead, rebar, wire rod, fuel oil, natural rubber and asphalt.
 - On February 28, 1993, Dalian Commodity Exchange was established, trading varieties of corn, yellow soybean No. 1, yellow soybean No. 2, soybean meal, soybean oil, palm oil, polypropylene, polyvinyl chloride, plastic, coke, coking coal, iron ore, plywood, fiberboard, eggs and other 15 futures varieties.
 - On March 1, 1993, Zhengzhou Grain Wholesale Market began to use the name of "China Zhengzhou Commodity Exchange".
 - On May 28, 1993, Zhengzhou Commodity Exchange officially launched futures trading. There are 16 futures varieties such as strong gluten wheat, common wheat, PTA, No.1 cotton, white sugar, rapeseed oil, early indica rice, glass, rapeseed meal and methanol.
 - In 1999, The State Council promulgated the Interim Regulations on the Administration of Futures Trading and four corresponding regulations on regulating futures exchanges, futures brokerage companies and their senior executives, which officially brought **China's futures market into the orbit of legal system.**

A brief history of futures products

- Futures are divided into: commodity futures, Treasury bond futures, stock index futures and so on.
- 2. Treasury bond futures
 - In December 1992, the Shanghai Stock Exchange was the first to open Treasury bond futures trading, launching a total of 12 varieties of Treasury bond futures contracts, only open to institutional investors.
 - On October 25, 1993, the Shanghai Stock Exchange opened Treasury bond futures trading to individual investors.
 - On February 23, 1995, Shanghai Wanguo Securities Company illegally traded the 327 contract, which became the "327 Treasury bond incident" that caused a sensation in the market.
 - On February 25, 1995, the CSRC and the Ministry of Finance jointly issued the Interim Measures for the Administration of Treasury Bond Futures Trading, which increased the ratio of trading margin and shrunk the trading venues from more than a dozen to four exchange markets.
 - **On May 17, 1995, the CSRC issued a notice and decided to suspend the trading of Treasury bond futures.** China's first trial of Treasury bond futures trading ended in failure.
 - On February 13, 2012, the network test of simulation trading of Treasury bond futures was launched. Eight financial institutions participated in the first round of network test, and then it was promoted to the whole market.
 - **On September 6, 2013, Treasury bond futures were officially re-listed and traded.**

A brief history of futures products

- Futures are divided into: commodity futures, Treasury bond futures, stock index futures and so on.
- 3. Stock index futures
 - On September 8, 2006, China Financial Futures Exchange (CFFEX) was established in Shanghai.
 - On October 30, 2006, CFFEX started the simulation trading of CSI 300 stock index futures.
 - On April 16, 2010, China's first stock index futures trading contract, CSI 300 stock index futures, was officially listed and traded in CFFEX.
 - On April 16, 2015, CFFEX launched the SSE 50 and CSI 500 stock index futures.
 - On July 22, 2022, the CSI 1000 stock index futures were officially listed in CFFEX.
- At present, there are four stock index futures products in China's futures market:
 - CSI300 沪深300股指期货
 - SSE50 上证50股指期货
 - CSI500 中证500股指期货
 - CSI1000 中证1000股指期货

Current status of China's futures market

- By the end of 2023, there were 120 varieties of futures and options listed on China's futures and stock exchanges, including 68 commodity futures, 8 financial futures, 32 commodity options and 12 financial options.
- In 2023, the trading volume of China's futures market was 8.501 billion lots with a turnover value of 568.5 trillion yuan. Among them, the trading volume of commodity futures and options accounted for 70% of the global total.
- **At present, the trading volume of China's commodity futures has ranked first in the world.**
 - In terms of metals, the Chinese market dominates the pricing of zinc futures and has an important influence on copper and aluminum futures.
 - In agricultural products, China dominates the pricing of sugar, soybeans, soybean meal and wheat, and has a significant influence on corn futures.

China's OTC financial derivatives market

- Over-the-counter or OTC financial derivatives market refers to the market for financial derivatives that are not traded on exchanges.
- OTC markets usually do not require standardized trading protocols or public quotes. Participants in OTC markets can trade more freely than they do on exchanges, but they also require more ability to manage risk.
- The trading objects of OTC financial derivatives market include various interest rates, stocks, exchange rates, commodity futures and other financial subjects, such as options, swaps, futures and so on.

Classification of OTC derivatives market

- By trading venues
 - Inter-bank OTC derivatives market
 - Bank-counter OTC derivatives market
 - Securities firms OTC derivatives market
- By traded products
 - Foreign exchange derivatives market
 - Interest rate derivatives market
 - Commodity derivatives market
 - Stock index derivatives market

History of OTC derivatives market in China

- 1997: The China inter-bank bond market was established, laying the foundation for financial derivatives trades.
- 2002: The **commercial bank counter** book-entry treasury bond trading business was launched.
- 2004: The State Administration of Foreign Exchange approved some banks to launch the forward foreign exchange business.
- 2005: Bond forward transactions were launched in the **inter-bank bond market**.
- 2006: The pilot of RMB interest rate swap transactions was launched, which was the first real interest rate derivative in the inter-bank market and promoted the process of interest rate liberalization.
- 2007: Credit risk mitigation tools (CRM), including credit risk mitigation agreements (CRMA) and credit risk mitigation warrants (CRMW), were created as the first credit derivatives launched in China's inter-bank market.
- 2012: The Securities Association of China released Regulations on the Counter Trading Business of Securities Companies, began the standardization of the securities company counter market and financial derivatives trading.
- 2013: The first batch of **securities companies of the counter market** obtained approval and launched financial derivatives such as income certificates and over-the-counter options.
- 2014: the Shanghai Clearing House officially launched the first batch of domestic OTC bulk commodity financial derivatives clearing services.
- 2015: With the abnormal fluctuations in the stock market, regulatory authorities strengthened the supervision of the derivatives business in the securities company counter market, standardized business operations.

History of OTC derivatives market in China

- 2022: The "**Futures and Derivatives Law**" was officially implemented, laying a solid legal foundation for the high-quality development of the futures and derivatives market.
- 2023: The "**Northbound Swap Connect**" was launched for trading, promoting the further development of the RMB internationalization and the inter-bank interest rate derivatives market.
 - The "Northbound Swap Connect" refers to the institutional arrangement through which overseas investors from Hong Kong, as well as other countries and regions, participate in the China's inter-bank financial derivatives market (primarily the interest rate swap market).
- In 2023, the notional principal of transactions in the inter-bank derivatives market reached 196.8 trillion yuan, with a year-on-year growth of 18.2%. The scale of the derivatives market grew steadily and the product series continued to be enriched.

Inter-bank OTC derivatives market

- It refers to OTC derivatives transactions that take place in the interbank market, where transactions take place between members of the interbank market.
- It is the most important component of OTC financial derivatives market in China.
- All kinds of institutions have different degrees of participation in the OTC derivatives market. Among them, commercial banks are the main body of OTC derivatives transactions in the interbank market, accounting for more than 90% of the total market trading volume.
- Main products of Inter-bank OTC derivatives market in China
 - exchange rate derivatives
 - interest rate derivatives
 - RMB foreign exchange swaps and interest rate swaps, they are the main trading varieties in the inter-bank OTC derivatives market.

Inter-bank OTC derivatives market

- In 2022, a total of 21.3 trillion yuan was traded in the interbank local currency derivatives market, basically flat year-on-year. Among them,
 - the total notional principal of interest rate swaps was 21.0 trillion yuan;
 - Standard bond forward transactions reached 260 billion yuan,
 - the notional principal of credit risk mitigation certificates was 26.8 billion yuan,
 - the notional principal of credit default swaps was 2.4 billion yuan.
- In 2022, Treasury bond futures totaled 46.4 trillion yuan, up 68.7 percent year on year.

Bank-counter OTC derivatives market

- It is a retail over-the-counter market where commercial banks trade with enterprises and individuals.
- Due to regulatory reasons, commercial banks are the main participants in China's OTC financial derivatives market. Securities companies, insurance companies and other financial institutions can only partially participate in the inter-bank OTC financial derivatives market, and have not yet handled OTC financial derivatives business for individuals.
- At present, there are interest rate swap, RMB foreign exchange forward, RMB foreign exchange swap, RMB foreign exchange currency swap, foreign exchange option and other products in China's over-the-counter derivatives market, which are mainly used to manage the risk of interest rate and exchange rate fluctuations.

Bank-counter OTC derivatives market

- According to data from the Balance of Payments report in 2019,
 - The total turnover of the foreign exchange forward market in 2019 was US \$380.6 billion
 - the forward transaction of foreign exchange by banks to customers carried out at the bank-counters totaled US \$304.7 billion;
 - the forward transaction of foreign exchange in the interbank foreign exchange market totaled US \$76 billion.
 - The total turnover of the foreign exchange and currency swap markets was US \$16.53 trillion
 - US \$16.41 trillion in interbank foreign exchange and currency swaps.
 - US \$119.5 billion in foreign exchange and currency swaps with customers at bank-counters.
 - The options market totaled \$850 billion
 - trading conducted by banks to customers was US \$268.8 billion at bank-counters.
 - interbank foreign exchange options market was US \$581.2 billion.
- Compared with the inter-bank OTC derivatives market, the trading venues of OTC derivatives market over the bank counter are scattered and the volume is small.
- ✓ The data of OTC derivatives business that commercial banks carried out with corporate clients over the counter is only reported to the CBRC and not disclosed to the market.

Securities firms OTC derivatives market

- The OTC derivatives business of securities companies relies on the counter system and quotation system of securities companies.
- The OTC derivatives that securities companies and futures risk management companies can engage in mainly include OTC options (场外期权) and total return swaps (收益互换).
 - Options business includes individual stock OTC options, index OTC options and commodity OTC options.
 - Swap business includes financing and margin lending.
- In 2022, the cumulative new nominal principal of OTC derivatives of securities companies was 8.31 trillion yuan; The nominal principal of the ending balance was 2.09 trillion yuan.
 - The total new nominal principal of OTC options was 3.35 trillion yuan; The nominal principal balance at the end of the period was 1.19 trillion yuan.
 - The total new nominal principal of return swaps was 4.96 trillion yuan; The nominal principal balance at the end of the period was 0.90 trillion yuan.

OTC Derivatives Market in China

- According to the Bank for International Settlements, the total notional amount of OTC derivatives in the world reached US \$598.4 trillion at the end of 2021.
- However, in 2021, the trading volume of OTC derivatives in China was only 2.33 trillion yuan.
- The trading volume of OTC derivatives in China was only 0.05% of the total notional amount of OTC derivatives in the world!

Development of China's options market

- Compared with the mature capital market, the development of Chinese option market started relatively late.
- At present in the exchange markets, the most active varieties are stock options and commodity options.
- The OTC market is dominated by foreign exchange options, and interest rate options are still in their infancy.
 - On the one hand, the formation of this pattern is related to the demand of investors. The bulk commodity trading is mainly dominated by enterprises and private equity funds, and the acceptance of option derivatives is high.
 - On the other hand, it is also affected by the development speed of spot and futures markets. At present, China's stock and commodity futures markets are relatively mature, and the corresponding supporting systems are relatively perfect, which promotes the development of related option markets.

Options in exchange markets of China

- 1. Stock Option
 - (1) Warrant 权证
 - In June 1992, the Shanghai Stock Exchange launched the Big Feile rights issue warrant
 - In June 1996, China Securities Regulatory Commission (CSRC) was forced to suspend trading of warrants in an emergency
 - On August 22, 2005, to promote the reform of non-tradable shares of listed companies in China, new warrants were set up, and the first one was Baosteel warrant.
 - **On August 12, 2011, the last warrant, Changhong subscription warrant, ended the transaction and the warrants have withdrawn from the historical stage.**
 - (2) Index fund options
 - In 2015, the Shanghai Stock Exchange launched SSE 50ETF options. China's capital market officially entered the option era.
 - In 2019, the Shanghai and Shenzhen stock exchanges respectively listed options on the CSI 300ETF.
 - (3) Stock index options
 - On December 23, 2019, the CSI 300 stock index options were listed on China Financial Futures Exchange. It is the first stock index option in China.
 - On July 22, 2022, the CSI 1000 stock index options were officially listed.
 - (4) Individual stock options
 - **At present China does NOT have a stock option (floor). here not to talk about OTC market.**

Options in exchange markets of China

- 2. Commodity Futures Option

- On March 31, 2017, soybean meal futures options were officially listed and traded on Dalian Commodity Exchange. China's officially entered the futures option era.
- On April 19, 2017, white sugar futures options were officially listed and traded on Zhengzhou Commodity Exchange.
- On September 21, 2018, copper futures options were officially listed and traded on the Shanghai Futures Exchange.
- On January 28, 2019, rubber futures options were officially listed and traded on the Shanghai Futures Exchange.
- On January 28, 2019, cotton futures options were officially listed and traded on Zhengzhou Commodity Exchange.
- On Jan 28, 2019, the corn futures option contract was listed and traded on the Dalian Commodity Exchange.
- On December 9, 2019, the iron ore futures option contract was listed and traded on the Dalian Commodity Exchange.

Over-the-Counter Options in China

- Foreign exchange options

- 1) Foreign exchange options are representative of OTC options
 - In April 2011, the interbank market launched RMB to foreign exchange options trading business.
 - Different from commodity options and stock options, which are centrally traded on major exchanges, foreign exchange options are mostly traded over the counter.
 - The currency pairs for China's foreign exchange option trading are RMB against US dollar, Hong Kong dollar, Japanese yen, Euro, British pound, Russian ruble and other currencies listed in the inter-bank foreign exchange market.
- 2) Foreign exchange options traded in the interbank wholesale market are mainly in short term.
 - The domestic RMB foreign exchange market is divided into two levels, namely,
 - the retail market between banks and customers (Enterprises and individual customers). The retail market of foreign exchange is widely distributed and scattered.
 - the inter-bank wholesale foreign exchange market (banks conduct foreign exchange transactions through the trading platform of the China Foreign Exchange Trade System, where banks balance their foreign exchange fund positions).
 - Compared with the two markets, the interbank foreign exchange market is larger and the corresponding foreign exchange option trading volume is higher.
 - In 2022, the annual options trading volume in the foreign exchange market was US \$1.45 trillion, including US \$293.3 billion in the client-to-client market and US \$1.24 trillion in the interbank market.

Over-the-Counter Options in China

- Interest rate options
 - With the approval of the People's Bank of China, the China Foreign Exchange Trade System started trial operation of interest rate option trading and related services from February 24, 2020.
 - In March 2020, the interest rate option business in the interbank market was officially put into operation, and the first batch of interest rate options linked to LPR (Loan Prime Rate, LPR) was launched. 111 financial institutions including commercial banks, securities companies and private equity funds, became the first batch of market participants.
 - The interest rate option trading varieties launched by the Foreign Exchange Trade Center are LPR1Y/LPR5Y interest rate swap options and interest rate upper/lower limit options, and the option type is European option. The trading method is over-the-counter trading.
 - The introduction of interest rate options linked to LPR can further meet the diversified interest rate risk management needs of financial institutions, and alleviate the impact of LPR fluctuations on banks' negative balance sheet.
 - So far, there is no publicly available trading data.

Regulators of China's derivatives market

- China Securities Regulatory Commission
- China Futures Association
- State Administration of Foreign Exchange
- People's Bank of China

Regulators of China's derivatives market

Exchange-Traded Market

- **Futures Market:**
 - Product Types: Commodity futures; Financial futures, including stock index futures, treasury bond futures, etc.
 - Regulatory Institutions:
 - **The China Securities Regulatory Commission (CSRC)** is the main regulatory body, responsible for formulating laws, regulations, and institutional rules for the futures market, and supervising market entities such as futures exchanges and futures companies.
 - **The China Futures Association**, as an industry self-regulatory organization, conducts self-regulation of the futures industry under the guidance of the CSRC, including formulating industry norms, conducting qualification examinations and training for practitioners, and carrying out self-inspection of member units.
- **Securities Exchange Derivatives Market:**
 - Product Types: Stock options, etc.
 - Regulatory Institutions:
 - **The China Securities Regulatory Commission (CSRC)** is responsible for overseeing the derivatives trading business of securities exchanges, including rule-making, product approval, and supervision of market trading behaviors.
 - **The Securities Association of China** assists the CSRC in self-regulation by formulating relevant business norms and guidelines to maintain market order.

Regulators of China's derivatives market

Over-the-Counter (OTC) Market

- **Interbank Derivatives Market:**
 - Product Types: Foreign exchange derivatives; Interest rate derivatives.
 - Regulatory Institutions:
 - **The State Administration of Foreign Exchange (SAFE)** is responsible for the supervision of foreign exchange derivatives businesses.
 - **The People's Bank of China** undertakes the responsibility of macro-prudential supervision, controls the stability of the financial system and financial markets, and issues relevant management policies.
 - **The China Banking and Insurance Regulatory Commission (CBIRC, now "NFRA")** conducts micro-prudential supervision on the participation of banking financial institutions in derivatives trading.
 - **The National Association of Financial Market Institutional Investors (NAFMII)**, as a self-regulatory organization, releases relevant agreements and rules to conduct self-regulation of the interbank derivatives market.
- **Securities Institution Counter Derivatives Market:**
 - Product Types: Personalized financial derivatives such as income certificates and over-the-counter options.
 - Regulatory Institutions:
 - **The China Securities Regulatory Commission** is responsible for formulating regulatory rules for the counter derivatives business of securities institutions, and managing aspects such as business access, daily supervision, and risk monitoring.
 - **The Securities Association of China** plays a role in self-regulation by formulating relevant business norms and guidelines to guide securities institutions to conduct business in compliance with regulations.

Problems in China's financial derivatives market

- 1. The development level of financial derivatives market lags behind and varieties are lacking.
 - There is a lack of subdivision of commodity futures varieties in China. The price discovery and hedging functions of futures market are difficult to cover all commodity market fields, and the needs of enterprises to predict the prices of more important commodities.
 - In the financial futures market, there are only stock index futures and treasury bond futures.

Problems in China's financial derivatives market

- 2. The lack of innovation of financial derivatives restricts the development of the market.
 - China's exchange market of derivatives still adopts a strict approval system, which reduces the operating efficiency, curtailing the trading demand and weakening the independent innovation of the exchange. In addition, regulators in China often design products from the perspective of convenient supervision, the real needs of market participants are ignored.
 - The credit risk mitigation (CRM) product, the Chinese version of CDS launched in 2010, is a typical case. International standard CDS products use a basket of debt with equal rights, while China's CRM products use designated debt. The design argument of the CRM product calls this transformation an innovation, and argues that it will be easier to deal with credit events, and it will be easier to judge the real trading intentions of market participants, which is conducive to market risk prevention. However, this killed the application value of CRM.

Problems in China's financial derivatives market

- 3. The financial derivatives market trading rules are not perfect, the international influence is low.
 - China's financial derivatives market lacks real market equilibrium price, information disclosure lacks transparency and openness, and the quality of participants in the financial derivatives market needs to be improved. In addition, only domestic investors are allowed to participate in the exchange financial derivatives market in China, and only some domestic financial institutions are allowed to participate in the OTC financial derivatives market.
 - The transaction prices of most financial derivatives have a weak influence on the pricing of international financial derivatives, and have little influence on the derivative markets of other countries and regions.

The trend of China's derivatives market

- Digitalized and intelligent development of derivatives market
- Increased openness to the outside world and internationalization
- Innovation and diversification of derivatives markets